Retail Sales Analytics — Code File

Prepared by: Ritwika Maji
Institution: Bankura Unnayani Institute of Engineering
Source file: retail_sales_analysis.py

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"""
Retail Sales Data Analytics and AI-Based Sales Prediction
Dataset: retail_sales_dataset.csv (synthetic sample data)
Run: pip install pandas numpy matplotlib scikit-learn
     python retail_sales_analysis.py
"""
from pathlib import Path
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score

DATA_FILE = Path(__file__).with_name("retail_sales_dataset.csv")
df = pd.read_csv(DATA_FILE)
df["Order_Date"] = pd.to_datetime(df["Order_Date"])
df["Month"] = df["Order_Date"].dt.month
df["DayOfWeek"] = df["Order_Date"].dt.dayofweek

print("\\n--- DATA OVERVIEW ---")
print(df.head())
print("\\nRows and columns:", df.shape)
print("\\nMissing values:\\n", df.isnull().sum())
print("\\nTotal sales: ■{:,.2f}".format(df["Sales_Amount"].sum()))
print("Average order sales: ■{:,.2f}".format(df["Sales_Amount"].mean()))
print("Units sold:", int(df["Units_Sold"].sum()))

print("\\n--- SALES BY CATEGORY ---")
print(df.groupby("Product_Category")["Sales_Amount"].sum().sort_values(ascending=False))
print("\\n--- SALES BY REGION ---")
print(df.groupby("Region")["Sales_Amount"].sum().sort_values(ascending=False))
print("\\n--- SALES BY CHANNEL ---")
print(df.groupby("Sales_Channel")["Sales_Amount"].sum().sort_values(ascending=False))

features = df[[
    "Product_Category", "Region", "Sales_Channel", "Units_Sold",
    "Unit_Price", "Discount_Percent", "Month", "DayOfWeek"
]]
target = df["Sales_Amount"]
features = pd.get_dummies(features, columns=[
    "Product_Category", "Region", "Sales_Channel"
], drop_first=False)

X_train, X_test, y_train, y_test = train_test_split(
    features, target, test_size=0.20, random_state=42
)
model = RandomForestRegressor(
    n_estimators=150, random_state=42, min_samples_leaf=2
)
model.fit(X_train, y_train)
predictions = model.predict(X_test)

mae = mean_absolute_error(y_test, predictions)
rmse = np.sqrt(mean_squared_error(y_test, predictions))
r2 = r2_score(y_test, predictions)
print("\\n--- MODEL EVALUATION (held-out test set) ---")
print(f"MAE: ■{mae:,.2f}")
print(f"RMSE: ■{rmse:,.2f}")
print(f"R²: {r2:.3f}")

# Export charts for the report/presentation
Path("outputs").mkdir(exist_ok=True)
df.groupby("Product_Category")["Sales_Amount"].sum().sort_values().plot(kind="bar")
plt.title("Total Sales by Product Category")
plt.ylabel("Sales Amount (■)")
plt.tight_layout()
plt.savefig("outputs/sales_by_category.png", dpi=150)
plt.close()

df.groupby("Region")["Sales_Amount"].sum().sort_values().plot(kind="bar")
plt.title("Total Sales by Region")
plt.ylabel("Sales Amount (■)")
plt.tight_layout()
plt.savefig("outputs/sales_by_region.png", dpi=150)
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plt.close()

plt.scatter(y_test, predictions, alpha=0.6)
plt.title("Actual vs Predicted Sales")
plt.xlabel("Actual Sales (■)")
plt.ylabel("Predicted Sales (■)")
plt.tight_layout()
plt.savefig("outputs/actual_vs_predicted.png", dpi=150)
plt.close()

print("\\nCharts saved in the outputs folder.")
```